EXP.NO: 4

Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.

DATE: 06/02/2025

AIM:

To design a web form that accepts the user's name and age, and to develop a Servlet program to process the form data and display it back on the browser.

ALGORITHM:

1. Input:

- The user enters their **name and age** in a web form.
- The form is submitted to the server.

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2. Form Creation:

1. Create an HTML form using `<form>` tag.
2. Add input fields:
    - Name (text)
    - Age (number)
3. Add a submit button.
4. Set:

```
action="NameAgeServlet"
method="post"
```

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3. Servlet Program Flow:

1. Create Servlet Class:

- Create a Java class extending `HttpServlet`.
- Use annotation:

```
@WebServlet("/NameAgeServlet")
```

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2. Handle Request:

- Override `doPost()` method.
- Retrieve values using:

```
request.getParameter("name");
request.getParameter("age");
```

---

3. Process Data:

- Store name and age in variables.
- Optionally validate age.

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4. Generate Response:

- Set content type:

```
response.setContentType("text/html");
```

- Use `PrintWriter` to display output:

```
Hello [name]! You are [age] years old.
```

---

4. Compile the Servlet:

- Compile using `javac`.
- Generate `.class` file.

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5. Deploy the Servlet:

- Place class file in:

```
WEB-INF/classes
```

- Configure using annotations or `web.xml`.

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6. Execute the Program:

- Start Apache Tomcat server.
- Open browser and access:

```
http://localhost:8080/your-project/index.html
```

---

7. Output Display:

- After submitting the form, the browser displays:

```
Hello [name]! You are [age] years old.
```

CODE:

Index.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>User Details</title>
</head>
<body>

<h2>Enter Your Details</h2>

<form action="NameAgeServlet" method="post">

    Name:
    <input type="text" name="name" required><br><br>

    Age:
    <input type="number" name="age" required><br><br>

    <input type="submit" value="Submit">

</form>

</body>
</html>
```

NameAgeServlet.java:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/NameAgeServlet")
public class NameAgeServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        String name = request.getParameter("name");
        String age = request.getParameter("age");
```

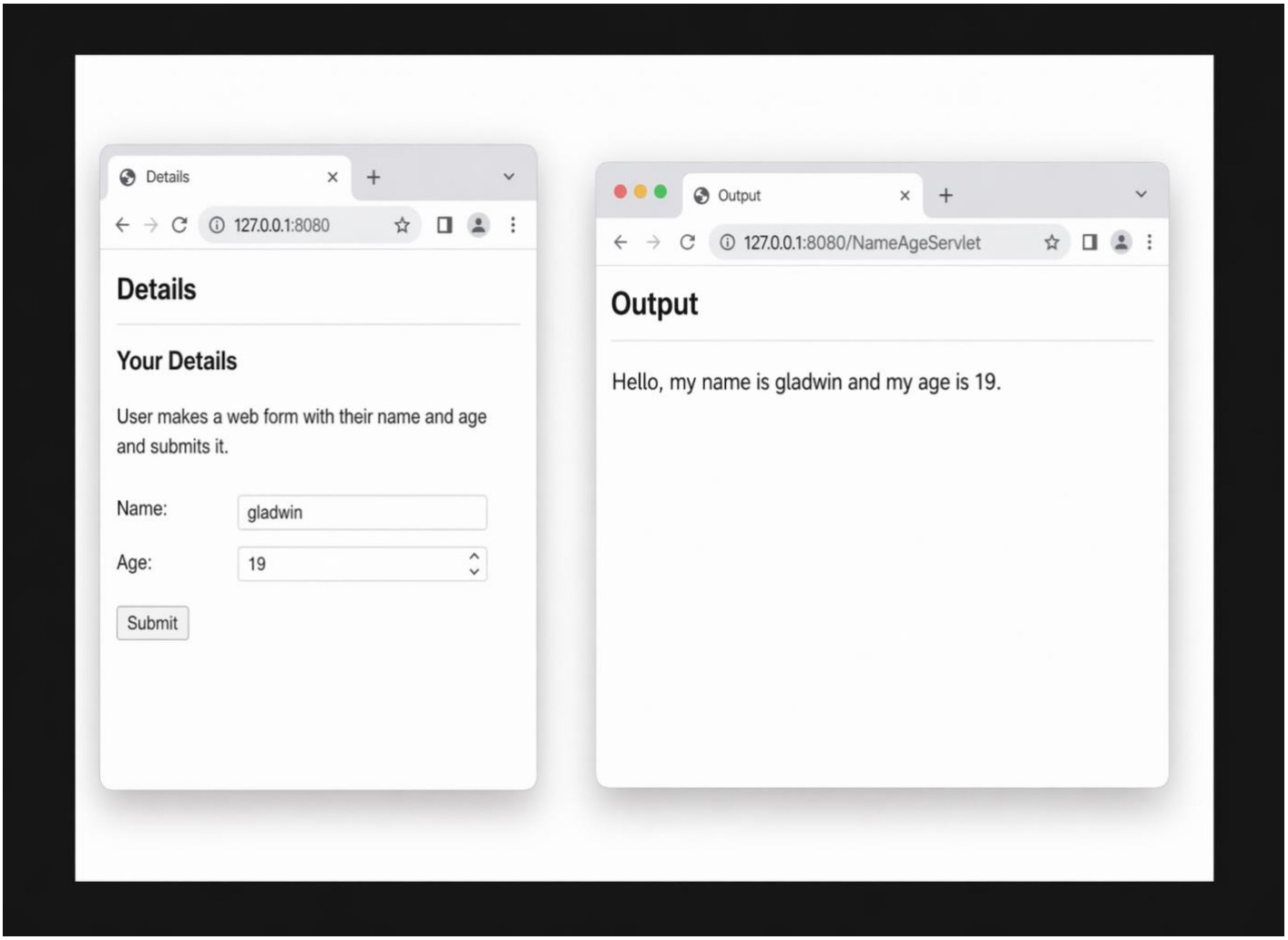
```
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<head><title>Output</title></head>");
        out.println("<body>");

        out.println("<h2>Hello, my name is " + name + " and my age is " +
age + ".</h2>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

OUTPUT:



RESULT:

The web form was successfully created and the Servlet program processed the user input and displayed the result on the browser.